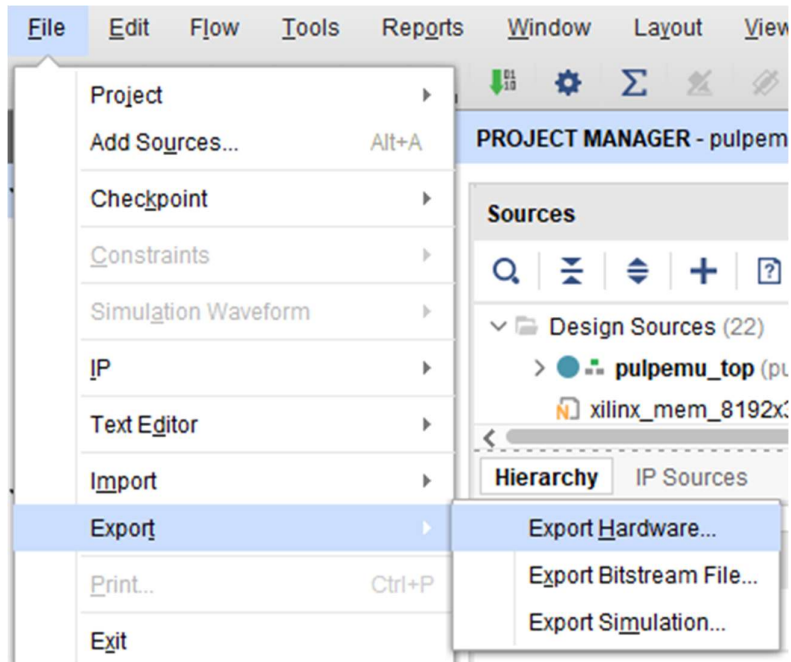
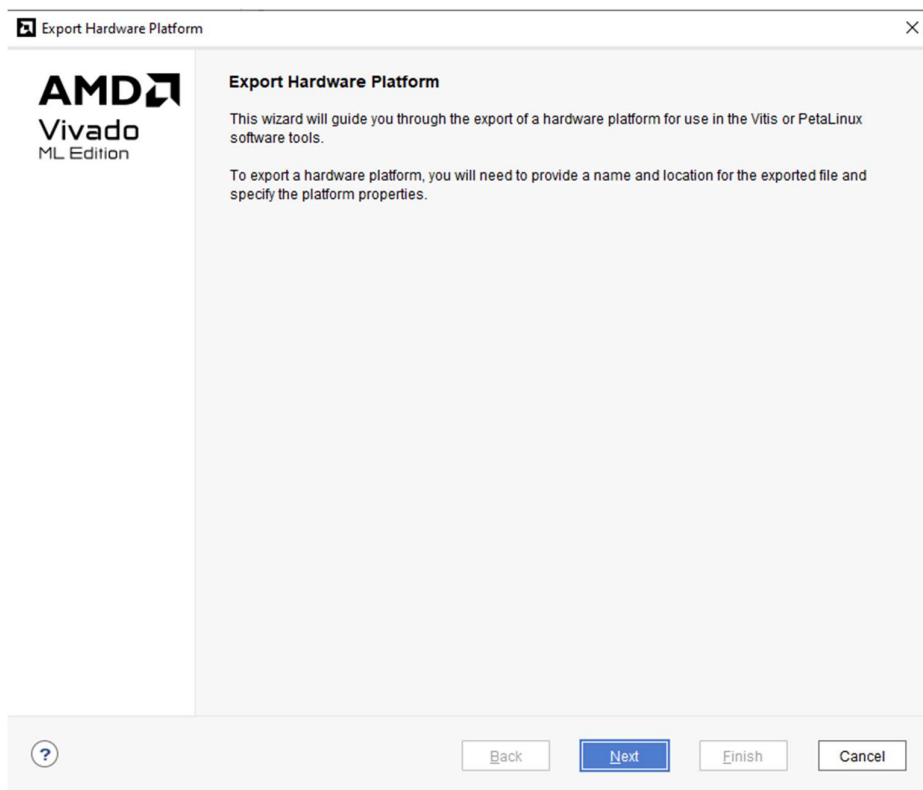


Bagian Vivado – Vitis (Proses pembuatan fsbl.elf dan BOOT.bin)

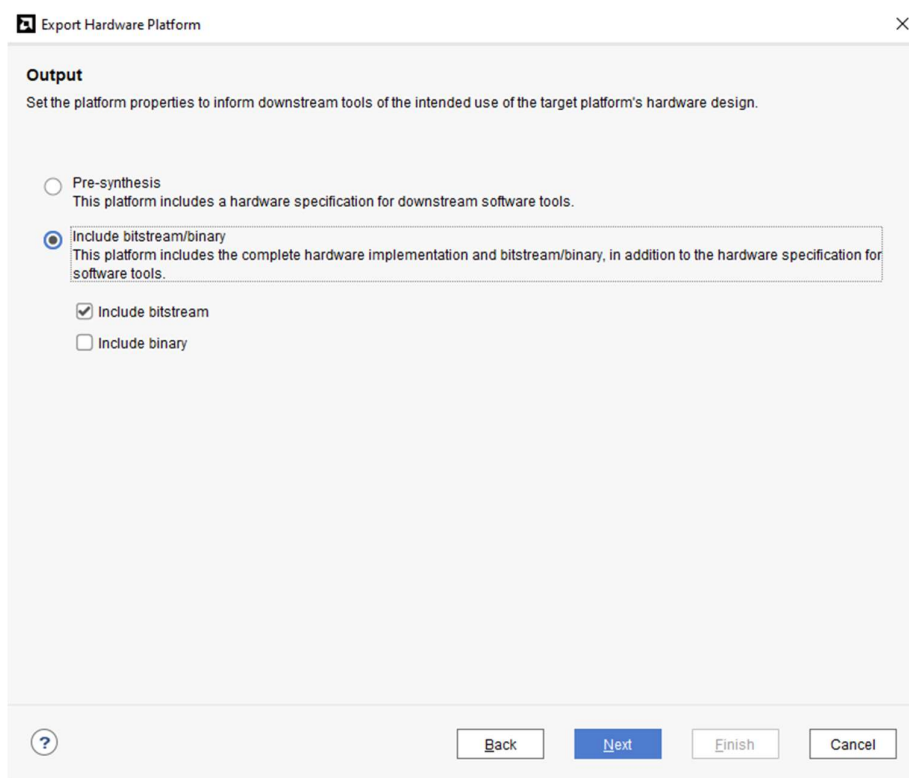
1. Buka pulpemu.xpr lalu export hardware



2. lalu Next



3. Centang include bitstream



Export Hardware Platform [X]

Output
Set the platform properties to inform downstream tools of the intended use of the target platform's hardware design.

☐ Pre-synthesis
This platform includes a hardware specification for downstream software tools.

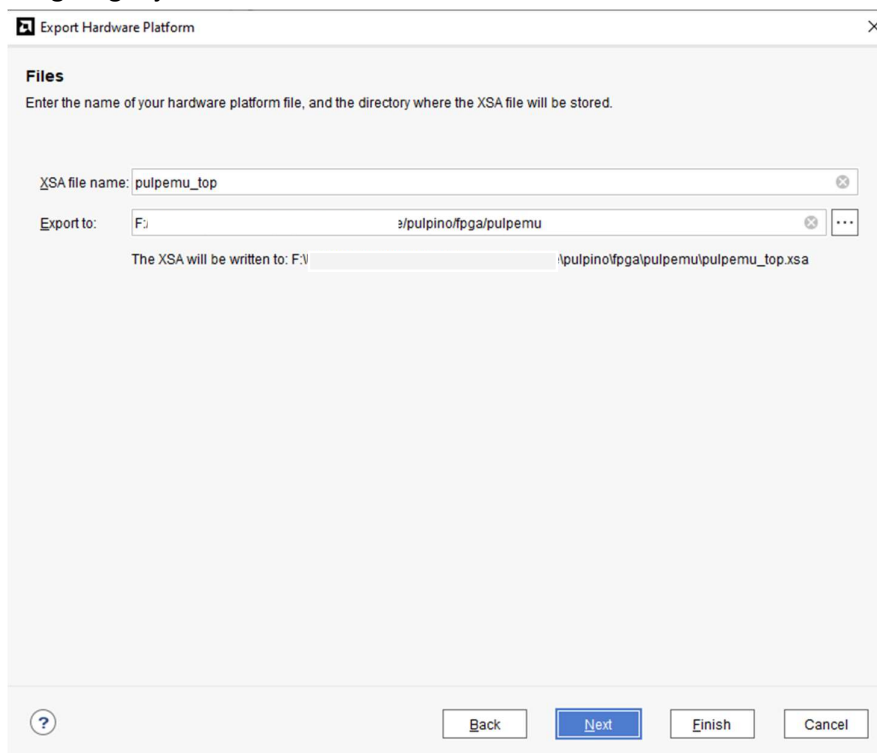
☒ Include bitstream/binary
This platform includes the complete hardware implementation and bitstream/binary, in addition to the hardware specification for software tools.

☒ Include bitstream

☐ Include binary

[?] [Back] [Next] [Finish] [Cancel]

4. Langsung saja next



Export Hardware Platform [X]

Files
Enter the name of your hardware platform file, and the directory where the XSA file will be stored.

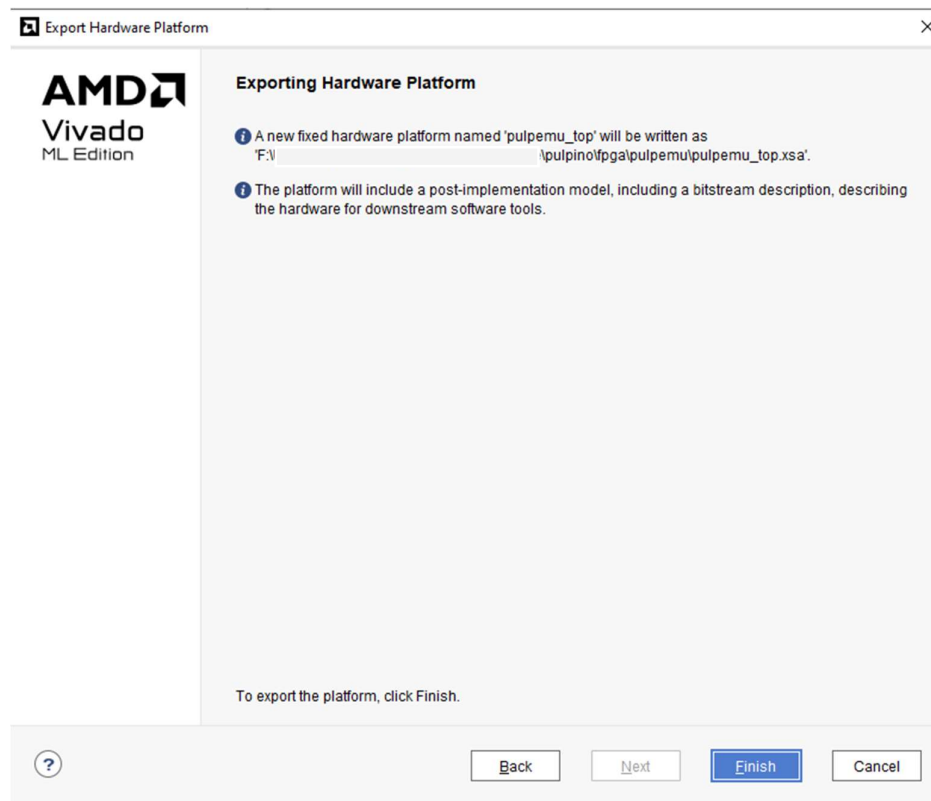
XSA file name:

Export to: [X] ...

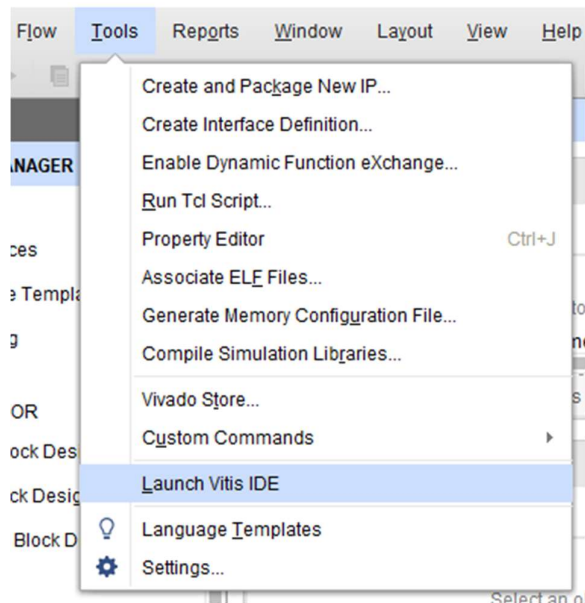
The XSA will be written to: F:\pulpino/fpga/pulpemu/pulpemu_top.xsa

[?] [Back] [Next] [Finish] [Cancel]

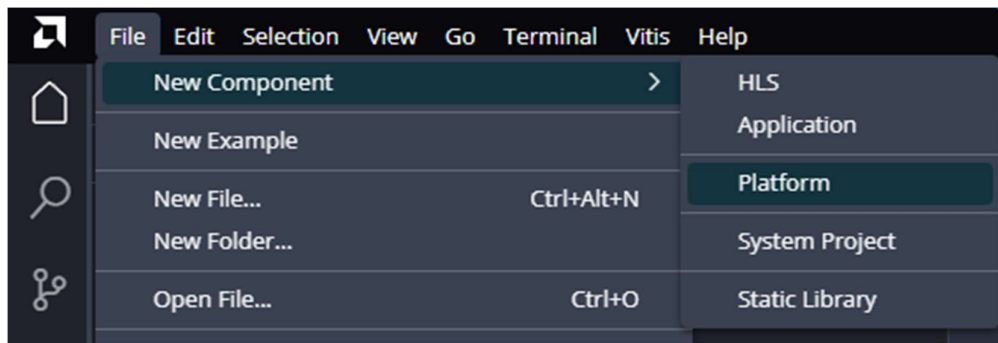
5. Ialu Next dan Finish



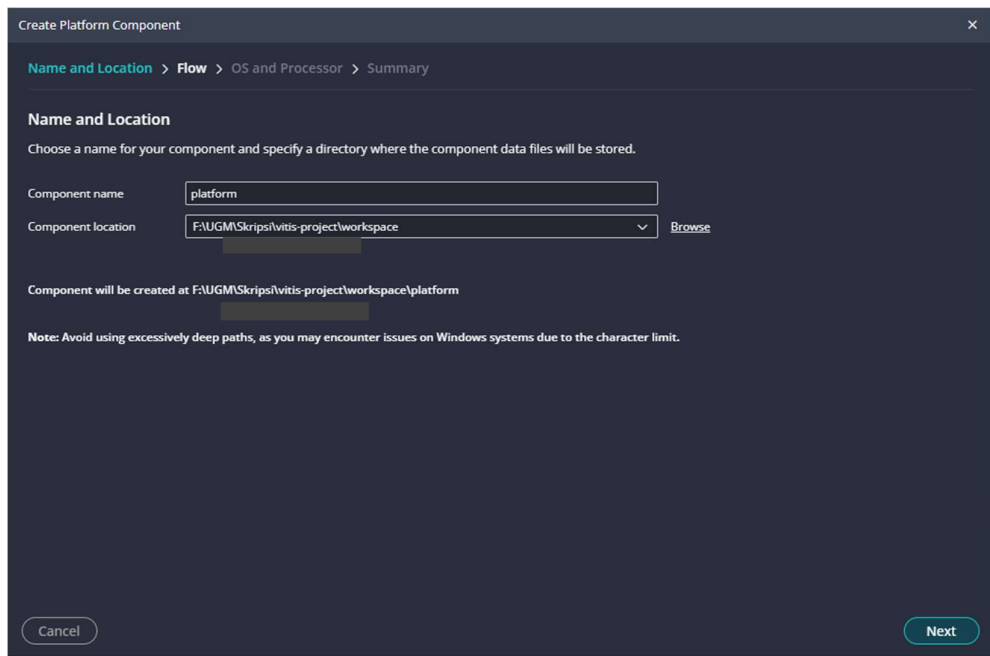
6. Setelah proses selesai, maka akan ada file .xsa di folder yang telah ditentukan sebelumnya
7. Buka aplikasi Vitis dari menu di Vivado



8. Pilih New Component > Platform



9. Atur Lokasi platform sesuai yang diinginkan



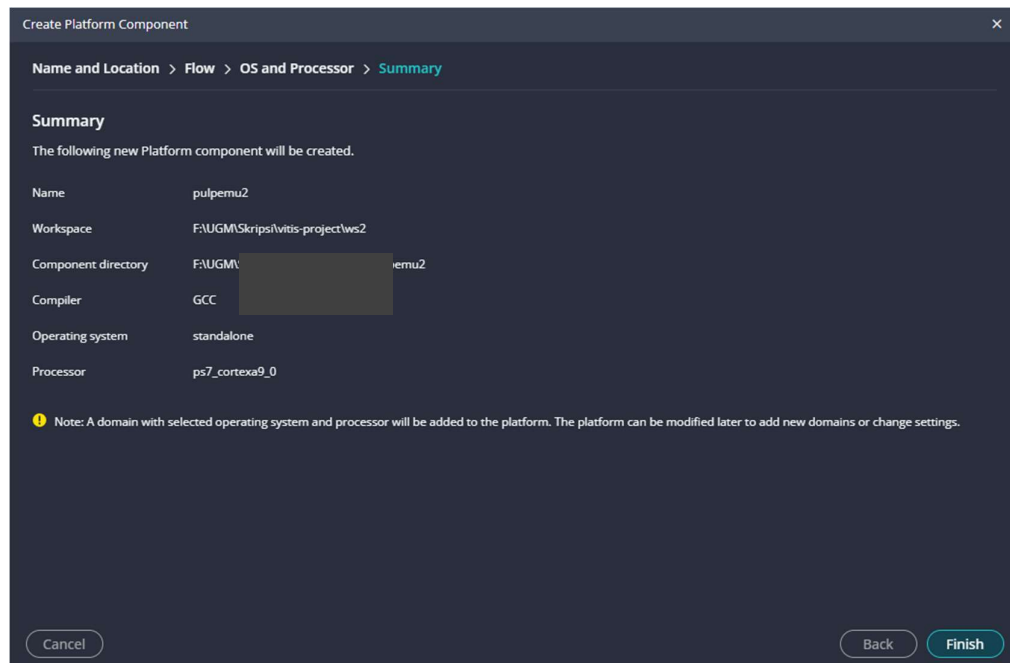
10. Pilih pulpemu_top.xsa yang sudah dibuat sebelumnya

The screenshot shows the 'Create Platform Component' dialog box with the 'OS and Processor' tab selected. The breadcrumb navigation is 'Name and Location > Flow > OS and Processor > Summary'. The section 'Select Platform Creation Flow' has two radio buttons: 'Hardware Design' (selected) and 'Existing Platform'. Below this, the 'Hardware Design (XSA) For Implementation' section shows a dropdown menu with 'F:\pino\pga\pulpemu...' selected and a 'Browse' button. The 'Advanced Options' section is expanded, showing four fields: 'SDT Source Repo' (empty), 'Board DTSI' (empty), 'User DTSI' (empty), and 'DT ZOCL' (checkbox, unchecked). Each of the first three fields has a 'Browse' button. At the bottom are 'Cancel', 'Back', and 'Next' buttons.

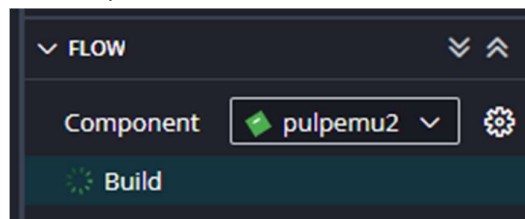
11. Tunggu sebentar maka akan terlihat seperti di bawah lalu Next

The screenshot shows the same 'Create Platform Component' dialog box, but now the 'OS and Processor' tab is selected. The breadcrumb navigation is 'Name and Location > Flow > OS and Processor > Summary'. The section 'Select Operating System and Processor' has a description: 'Specify the details for the initial domain to be added to the platform component. The platform can be modified later to add new domains or change settings by opening the platform editor.' Below this are three dropdown menus: 'Operating system' (selected: 'standalone'), 'Processor' (selected: 'ps7_cortexa9_0'), and 'Compiler' (selected: 'gcc'). There is a checked checkbox for 'Generate Boot artifacts'. At the bottom are 'Cancel', 'Back', and 'Next' buttons.

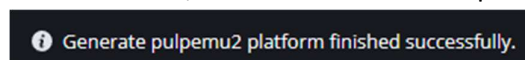
12. Lalu Next



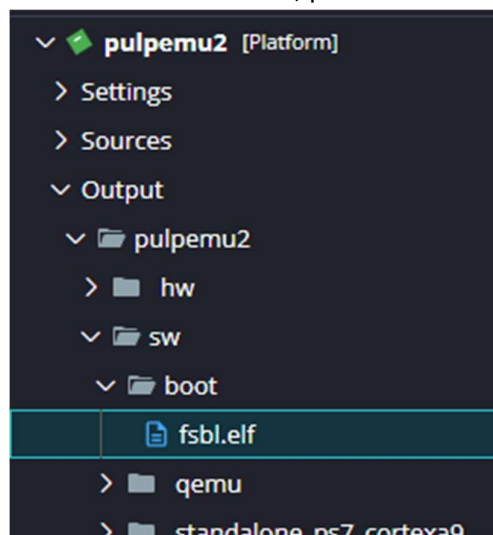
13. Setelah platform berhasil dibuat, maka lakukan build



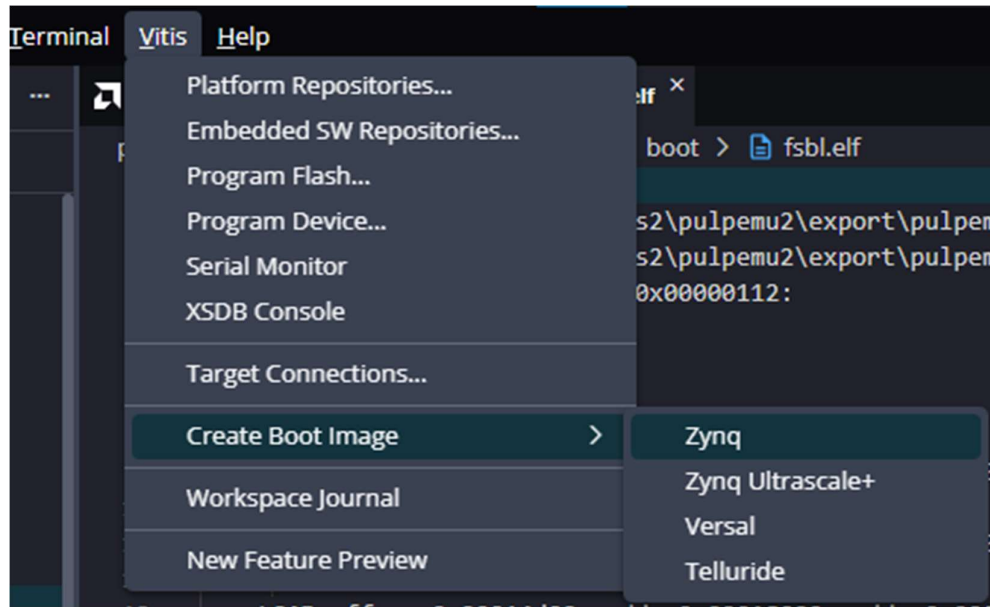
14. Setelah sukses, maka akan bisa dilihat pada notifikasi sebelah kanan bawah



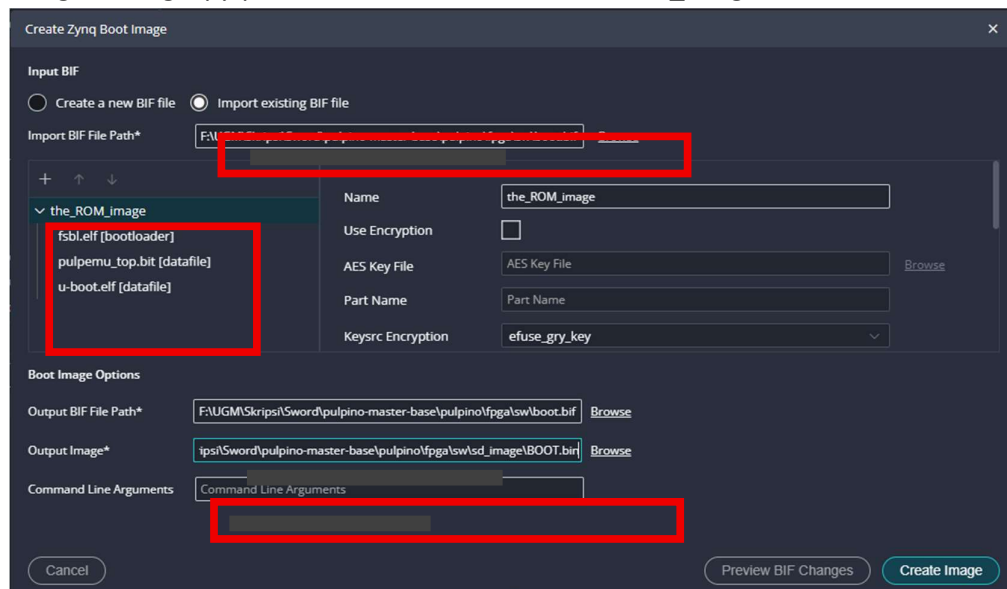
15. Setelah berhasil dibuild, pastikan file fsbl.elf ini tersedia



16. Langkah selanjutnya adalah membuat file boot



17. Import file boot.bif yang ada di fpga/sw. Pastikan juga file fsbl.elf, pulpemu_top.bit, dan u-boot.elf tersedia. Bisa dengan menyesuaikan path file terkait, atau bisa juga dengan mengcopy-paste file terkait ke dalam folder sd_image



18. Berhasil

